

Medicaid Opioid Prescribing Mapping Tool
Methodology
May 2024

Background

The Medicaid State Opioid Prescribing Mapping Tool is an online interactive mapping tool that presents geographic comparisons of the opioid prescribing rate at the state, county, and ZIP code levels. This mapping tool allows the user to see both the number and percentage of de-identified Medicaid opioid claims to better understand how this critical issue impacts states nationwide.

Opioid prescribing rates are derived using Medicaid data on prescription drugs prescribed by health care providers and reported by states to the Centers for Medicare & Medicaid Services (CMS). The mapping tool presents data for the most recent data year as well as one- and five-year change rates. For example, the release of 2019 data included changes in rates from 2018 to 2019 and changes in rates from 2014 to 2019. The underlying data tables are available for each calendar year, 2013 through the most recent data year for national- and state-level data, and for 2015 through the most recent data year for county- and zip-level data. The mapping tool does not contain beneficiary information.

Methods

The national and state measures presented in this mapping tool were aggregated from the Medicaid State Drug Utilization Data¹ (excluding over-the-counter prescriptions). Details about the Medicaid State Drug Utilization Data collected through the Medicaid Drug Rebate Program are publicly available.² These data include state-level reports listing the number of prescription fills and amounts paid by states by National Drug Code (NDC). County- and zip-level measures presented in this tool were aggregated from the Transformed Medicaid Statistical Information System (T-MSIS) Analytic Files (TAF). In order to be included in this tool, state-level TAF data had to meet quality checks for each data year. Due to known data quality issues in prior years, TAF data is presented only for data years 2016 and onward. More information about TAF data quality can be found in the Medicaid Data Quality Atlas.³ Data were summarized by NDC and linked to a list of NDCs classified as opioids.

In order to keep consistent reporting across the levels of geography, we develop a ZIP code reference table to determine the beneficiary's county. The ZIP code associated with the beneficiary is used to determine the county. For additional information on the methods used in developing the reference table, see Appendix A, "ZIP Code Reference Table Methodology".

The list of opioids included in the Opioid Prescribing Mapping Tool is generated using the Medicare Part D Overutilization Monitoring System (OMS), which can change from year to year. Opioid claims for each calendar year are identified using the OMS opioid list updated each year.⁴ Long-acting opioid claims are identified using

¹ <https://www.medicaid.gov/medicaid/prescription-drugs/state-drug-utilization-data/index.html>

² <https://www.medicaid.gov/medicaid/prescription-drugs/medicaid-drug-rebate-program/index.html>

³ <https://www.medicaid.gov/dq-atlas/welcome>

⁴ For additional information on Medicare Part D OMS please see <https://www.cms.gov/Medicare/Prescription-Drug-Coverage/PrescriptionDrugCovContra/RxUtilization.html>.

controlled substances including opioids with oral morphine milligram equivalent (MME) data made public by the Centers for Disease Control and Prevention (CDC)⁵.

Fee-for-service claims indicate the drug was prescribed in a Fee-For-Service (FFS) program and Managed Care claims indicates the drug was prescribed in a Managed Care Organization (MCO).

Data Limitations

In order to protect the privacy of beneficiaries, suppression is applied to the Opioid Prescribing Mapping Tool data. At each aggregated geographic level, any derived opioid claim counts between 1 and 10 are suppressed (replaced with missing values). Secondary suppression is applied in cases where a single subgroup category is primary suppressed. For example, if the Fee-for-Service opioid claim count for a given state is primary suppressed but the Managed Care opioid claim count is not primary suppressed, then the Managed Care opioid claim count for that state must be secondary suppressed to prevent disclosure of the Fee-for-Service opioid claim count suppressed value. Secondary suppression prevents backing into a primary suppressed value by using the values from the other opioid claim counts such as total opioid claim count.

The information presented in this mapping tool does not indicate the quality or appropriateness of care provided by individual physicians or in a given geographic region. For instance, high opioid prescribing for beneficiaries with cancer, palliative care and end of life care may be appropriate. In addition, opioids such as Methadone, may be prescribed for pain management as well as part of drug addiction detoxification and maintenance program. The mapping tool does not distinguish the various reasons for opioid prescribing.

Finally, the number of claims reflect prescriptions that have been filled by the beneficiary. The mapping tool does not measure whether a beneficiary took the medication.

How to Use the Mapping Tool

The Medicaid Opioid Prescribing Mapping Tool allows the user to explore opioid prescribing rates and changes in rates at the state level. The user can navigate around the map by clicking and dragging to a point of interest, as well as zooming in or out to adjust the state map to a larger or smaller view. Zooming in or out can be done by scrolling the mouse wheel, typing the “+” or “-” keys on the keyboard, or by clicking the “+” or “-” buttons on the mapping tool. The user can obtain more information about a particular state by clicking on the map, where a pop-up box displays opioid prescribing rates as well as the number of opioid claims and overall claims. For the maps that display rates by plan type, the user can click on the pop-up box and then move the slider bar to compare Fee-for-Service opioid prescribing rates to Managed Care opioid prescribing rates. Fee-for-Service opioid prescribing rates are shown to the left of the slider bar and Managed Care opioid prescribing rates are shown to the right of the slider bar. The web browsers that currently support the mapping tool include: Edge, Firefox, Chrome, Safari, iOS Safari, Chrome for Android.

The data for the mapping tool can be downloaded [here](#).

⁵ This information is available for download on the CDC Opioid Overdose Data Resources web page at: <https://www.cdc.gov/drugoverdose/resources/data.html>

CMS is obligated by the federal Privacy Act, 5 U.S.C. Section. 552a and the HIPAA Privacy Rule, 45 C.F.R Parts 160 and 164, to protect the privacy of individual beneficiaries and other persons. All direct identifiers have been removed from this data file.

Appendix A ZIP Code Reference Table Methodology

The ZIP code reference table contains a crosswalk between ZIP codes and FIPS state and county codes, FIPS-based geographic data elements (e.g., Core Based Statistical Areas), census-tract based Rural-Urban Area Commuting (RUCA) codes, and ZIP-code based health care regions (i.e., Dartmouth Atlas Hospital Referral Regions and Hospital Service Areas).

To construct the crosswalk between ZIP and FIPS state and county codes, we use the U.S. Department of Housing and Urban Development (HUD) ZIP-FIPS crosswalks.⁶ The HUD crosswalks report the proportion of a ZIP code's addresses that are contained within a particular FIPS code by quarter. We take these quarterly crosswalks and create an annual file that contains the last-best quarterly record for each ZIP-FIPS code combination within the year. Since ZIP codes can span multiple counties, the annual crosswalk contains multiple ZIP-FIPS value combinations. We use the ZIP-FIPS address proportions to assign a ZIP code to a single FIPS code by selecting the FIPS code that has the highest proportion of all the ZIP code's addresses. We then merge on FIPS-based geographic data elements to the ZIP reference table by FIPS.

We use a similar process to link ZIP codes to RUCA codes. RUCA codes classify U.S. census tracts using measures of population density, urbanization, and daily commuting.⁷ Researchers often use these codes to designate geographic areas as either urban or rural based on their commuting patterns to metropolitan and micropolitan areas. To link RUCAs to ZIP codes, we use the HUD ZIP-census tract crosswalks (ZIP-TRACT). As with the ZIP-FIPS crosswalks, the ZIP-TRACT crosswalks report the proportion of a ZIP code's addresses that exist in a given census tract by quarter. We use a similar process of identifying the census tract with the highest proportion of a ZIP code's addresses to create a unique ZIP-census tract combination for the reference table. We then merge RUCA codes onto the ZIP reference table by census tract.

While the HUD crosswalks are the main data source for the ZIP code reference table, we do perform additional edits to correct and enhance these data. Some records in the HUD ZIP-FIPS crosswalk contain FIPS values that are out of date. We correct these FIPS values using the closest annual ZIP-FIPS crosswalk file with correct FIPS values for the ZIP code. Additionally, some ZIP-TRACT quarterly crosswalks exclude ZIP codes that appear in the ZIP-FIPS crosswalk for the same year. While ZIP codes can change over time, we would not expect the presence of ZIP codes to differ between the two crosswalks within a given year, particularly since census tracts only change with the decennial census. To correct this discrepancy, we impute ZIP-TRACT crosswalk records from the closest annual ZIP-TRACT crosswalk file that contains the missing ZIP codes. Lastly, since we use the ZIP reference table to assign geographic data elements to health care providers based on their ZIP code, we supplement the ZIP crosswalk reference table with business ZIP codes that appear in the U.S. Census County Business Patterns (CBP) data.⁸ These ZIP codes represent codes for business campuses that may not appear in databases of general ZIP codes.

⁶ https://www.huduser.gov/portal/datasets/usps_crosswalk.html

⁷ <https://www.ers.usda.gov/data-products/rural-urban-commuting-area-codes.aspx>

⁸ <https://www.census.gov/programs-surveys/cbp/data.html>